XMM-Newton Observation of the AGN 3C 111

Part C: X-ray Spectral Analysis with SAS and XSPEC

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Abstract

I present the spectral analysis of the XMM-Newton EPIC-pn observation of the broad-line radio galaxy 3C 111 (ObsID 0552180101). Starting from the cleaned event list, good time intervals and extraction regions of Part A, source and background spectra were extracted with SAS 22.1.0, and redistribution (RMF) and ancillary (ARF) response files were generated. The source has a net count rate of $8.19 \pm 0.01\,\mathrm{counts\,s^{-1}}$ in a livetime of 42.2 ks, with the background contributing less than 1 per cent and no significant pile-up. The 0.3–10 keV spectrum was fitted in XSPEC. An absorbed power law gives $N_{\mathrm{H}} = (1.054 \pm 0.009) \times 10^{22}\,\mathrm{cm^{-2}}$ and $\Gamma = 1.627 \pm 0.008$, but leaves a clear excess near 6.1 keV (observed frame). Adding a Gaussian emission line improves the fit by $\Delta\chi^2 = 45.6$ for three additional parameters; the line has a rest-frame energy of $6.38 \pm 0.05\,\mathrm{keV}$, consistent with neutral Fe Kα, and an equivalent width of $41 \pm 12\,\mathrm{eV}$. For this best-fitting model the observed and unabsorbed 2–10 keV fluxes are 4.59×10^{-11} and $4.86 \times 10^{-11}\,\mathrm{erg\,cm^{-2}\,s^{-1}}$, and the rest-frame 2–10 keV luminosity is $(2.70 \pm 0.01) \times 10^{44}\,\mathrm{erg\,s^{-1}}$. The residual excess in χ^2 is confined below 3 keV and is attributed to calibration effects at the high count rate; the 3–10 keV band alone is fitted with $\chi^2/\mathrm{dof} = 108.6/106$.

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1 Introduction

The X-ray spectrum of an active galactic nucleus (AGN) is, to first order, a power law $N(E) \propto E^{-\Gamma}$, produced when ultraviolet photons from the accretion disc are Compton up-scattered by a hot corona of electrons above the disc. The photon index Γ encodes the temperature and optical depth of the corona. Along its way to the observer the power law is modified by photoelectric absorption in cold gas, both in the host galaxy and in the Milky Way, which removes soft X-rays and is characterised by the equivalent hydrogen column density $N_{\rm H}$. Part of the primary continuum is also reprocessed by surrounding matter, producing a fluorescent Fe Kα emission line at 6.4 keV (rest frame) and a reflection continuum. Fitting these components to the observed spectrum yields the intrinsic properties of the source.

3C 111 is a nearby ($z = 0.049$) broad-line radio galaxy with a powerful one-sided relativistic jet, located at low Galactic latitude behind a large column of Galactic gas. Its X-ray spectrum is therefore expected to be strongly absorbed at low energies.

Objective of Part C. Extract the source and background spectra from the cleaned EPIC-pn data, generate the response files (RMF and ARF), plot the spectrum, fit it with an absorbed X-ray spectral model in XSPEC, and determine the best-fit spectral parameters: hydrogen column density $N_{\rm H}$, photon index Γ, normalisation, observed and unabsorbed fluxes, X-ray luminosity, flux density, and the fit statistic.

2 Input Data and Software

The analysis starts from the products of Part A (Table 1). Table 2 lists the software used.

Table 1: Input products from Part A.

Clean event list	`pn_clean.fits` (461 931 events, CCD 4)
Good time intervals	extension `GTI00003` of `pn_clean.fits`, 20 intervals, 60.3 ks
Observing mode	EPIC-pn PrimeSmallWindow, Thin1 filter
Source region	circle, centre $(X, Y) = (27308.7, 27578.4)$, $r = 30''$ (600 sky units)
Background region	circle, centre $(X, Y) = (25689, 24313)$, $r = 40''$ (800 sky units)
Calibration index	`ccf.cif` (129 CCF entries)

Table 2: Software used.

SAS	22.1.0 (Ubuntu 24.04 build)
XSPEC	13 (HEASoft)
Python	3.12 with `astropy`, `numpy`, `matplotlib`
Plotting	`matplotlib`; `pymupdf` to read the `epatplot` output

Calibration files. The SAS response tasks read many Current Calibration Files (CCFs) that were not needed in Part A. The missing EPIC-pn, mirror (`XRT3_XAREAEF`, `XRT3_XENCIREN`) and general XMM CCFs listed in `ccf.cif` were downloaded from the HEASARC mirror before the responses could be generated. Because the ODF summary file copied into the working directory was truncated and could not be re-ingested by `cifbuild`, the calibration index of Part A was reused; it was built for the observation date and is therefore valid.

3 Spectral Extraction

3.1 Source and background spectra

Spectra were extracted with `evselect` from the clean event list, using the regions and good time intervals of Part A. Only events from CCD 4 with `FLAG==0` and `PATTERN<=4` (singles and doubles) were accepted. The spectra cover the standard pn channel range 0–20479 with a bin size of 5:

```
evselect table=pn_clean.fits:EVENTS withspectrumset=yes spectrumset=pn_src_spec.pi \
  energycolumn=PI spectralbinsize=5 withspecranges=yes specchannelmin=0 specchannelmax=20479 \
  expression="(CCDNR==4)&&(FLAG==0)&&(PATTERN<=4)&&((X,Y) IN circle(27308.7,27578.4,600))&&gti(
    ↪ pn_clean.fits:GTI00003,TIME)"
```

The background spectrum was extracted with the same command, using `circle(25689,24313,800)` and `spectrumset=pn_bkg_spec.pi`. The source region contains 348 014 events and the background region 3 579 events.

3.2 Area scaling

The task `backscale` computes the effective extraction area of each region, taking bad pixels and chip gaps into account, and writes it to the `BACKSCAL` keyword:

```
backscale spectrumset=pn_src_spec.pi badpixlocation=pn_clean.fits
backscale spectrumset=pn_bkg_spec.pi badpixlocation=pn_clean.fits
```

The values are 1 132 600 (source) and 2 010 700 (background); their ratio, 0.563, equals $(30/40)^2$ as expected for two unobstructed circles. Scaled to the source area, the background contributes about 2 000 counts, or 0.6 per cent of the source spectrum. Background subtraction therefore has a negligible effect on the results.

3.3 Response files

The redistribution matrix file (RMF) describes how a photon of a given energy is distributed over detector channels; the ancillary response file (ARF) gives the effective area as a function of energy, including mirror area, filter transmission, quantum efficiency and the fraction of the point-spread function (PSF) enclosed by the source region:

```
rmfgen spectrumset=pn_src_spec.pi rmfset=pn_src.rmf withenergybins=yes \
  energymin=0.1 energymax=12 nenergybins=2400
arfgen spectrumset=pn_src_spec.pi arfset=pn_src.arf withrmfset=yes rmfset=pn_src.rmf \
  badpixlocation=pn_clean.fits detmaptype=psf
```

The RMF has 2400 energy bins between 0.1 and 12 keV. The ARF was computed for a point source at the off-axis angle of 3C 111 ($\theta = 67.9'$), using the elliptical-beta PSF model; the fraction of the source region lost to bad pixels and chip gaps is negligible. Because the source region ($r = 30''$) does not enclose the whole PSF, the ARF corrects the flux for the missing PSF wings.

3.4 Grouping

To make χ^2 statistics valid, the spectrum was grouped with `specgroup` to at least 25 counts per bin, without oversampling the instrumental energy resolution by more than a factor of 3:

```
specgroup spectrumset=pn_src_spec.pi mincounts=25 oversample=3 rmfset=pn_src.rmf \
  arfset=pn_src.arf backgndset=pn_bkg_spec.pi groupedset=pn_src_grp.pi
```

The grouped spectrum `pn_src_grp.pi` has 218 groups, of which 177 lie in the fitted 0.3–10 keV band. The background, RMF and ARF file names are stored in its header keywords `BACKFILE`, `RESPFILE` and `ANCRFILE`, so XSPEC loads them automatically.

3.5 Pile-up check

The source count rate is about $8\,\mathrm{counts\,s^{-1}}$, so I checked for pile-up before fitting. When two photons land in the same frame close to each other, they are read out as one event with a larger pattern. Pile-up therefore lowers the fraction of single events and raises the fraction of doubles compared with what is expected.

I ran `epatplot` on the source events (Fig. 1a). In 0.5–2 keV it gives observed-to-model ratios of $s = 1.026 \pm 0.004$ for singles and $d = 1.014 \pm 0.006$ for doubles. Both are above 1, which at first looks odd since the fractions have to add up to one. The reason is that the clean event list from Part A only has `PATTERN<=4`. There are no triples or quadruples left in the data (their observed fractions in Fig. 1a are zero), but the `epatplot` model still expects about 2 per cent of them. So the singles and doubles come out slightly high in both cases.

To compare like with like, I read the observed and expected fractions from the `epatplot` output and renormalised the expected curves to singles plus doubles only (Fig. 1b). With this, the observed fractions follow the expected ones over the full energy range. In 0.5–2 keV the ratios are 1.015 for singles and 0.96 for doubles. This is a small deficit of doubles, which is the opposite of what pile-up would do. Part A ran `epatplot` on the unscreened events with all patterns and got $s = 1.003 \pm 0.004$ and $d = 0.991 \pm 0.005$. I conclude that the spectrum is not piled up, and I used the full $30''$ source circle. The small shift from doubles to singles is discussed again in Section 4.6.

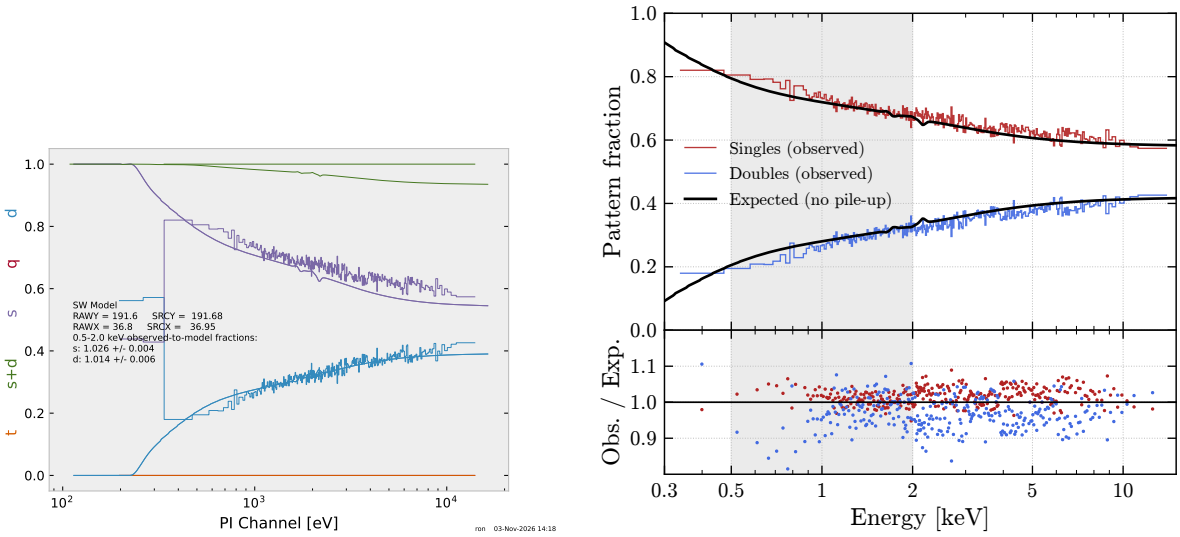
3.6 Extracted spectrum

The net (background-subtracted) count rate is $8.192 \pm 0.014\,\mathrm{counts\,s^{-1}}$ in a livetime of 42 234 s, corresponding to about 3.5×10^5 net source counts. The livetime is shorter than the 60.3 ks good-time interval because of the Small Window dead time (livetime fraction ≈71 per cent). Fig. 2 shows the spectrum. The strong low-energy cut-off below ∼1 keV is produced by photoelectric absorption combined with the falling effective area; the sharp drops near 1.8 and 2.2 keV are instrumental edges (Si K and the Au M edges of the mirror coating) that are folded into the response. A weak excess is visible near 6.1 keV.

4 Spectral Fitting

4.1 Method

Data products were made with SAS (Gabriel et al., 2004). All fits were performed in XSPEC (Arnaud, 1996) over 0.3–10 keV with χ^2 statistics. Absorption was described with the Tübingen–Boulder model `tbabs` with the abundances of Wilms et al. (2000) and the cross-sections of Verner et al. (1996). A single absorber was used for the Galactic and any intrinsic absorption, since the two cannot be separated at the resolution of the pn camera. The continuum was modelled with `zpowerlw`, a power law at the redshift of 3C 111 ($z = 0.049$, frozen). Errors are 90 per cent confidence intervals for one parameter of interest ($\Delta\chi^2 = 2.706$). Because the reduced χ^2



(a) (b)

Figure 1: Pile-up check for the source region. (a) Output of `epatplot`. The histograms are the observed pattern fractions and the smooth curves are the fractions expected without pile-up, for singles (s), doubles (d), triples (t), quadruples (q) and singles plus doubles (s+d). (b) Observed single and double fractions compared with the expected fractions renormalised to singles plus doubles, since the clean event list has only `PATTERN<=4`. The lower panel shows the ratio of observed to expected fractions. The grey band is 0.5–2 keV, the range used by `epatplot` for the pile-up test.

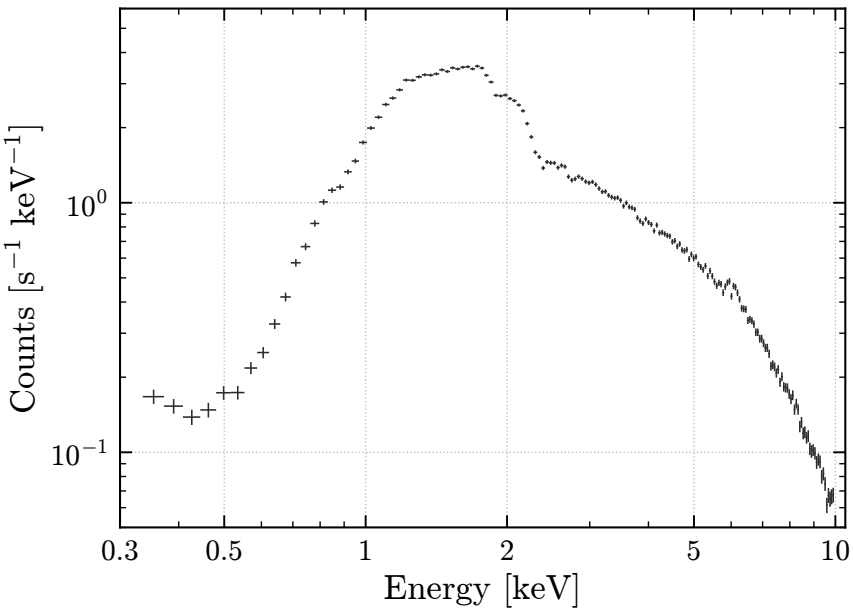


Figure 2: Background-subtracted XMM-Newton EPIC-pn spectrum of 3C 111 (ObsID 0552180101), grouped to a minimum of 25 counts per bin.

exceeds 2, XSPEC's default limit for error calculation was raised with `error maximum 10.0`. A flat ΛCDM cosmology with $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_\Lambda = 0.73$ was adopted (`cosmo 70 0 0.73`), giving a luminosity distance $D_L = 217.9\,\mathrm{Mpc}$.

The complete XSPEC script is given in Appendix B.

4.2 Absorbed power law

The first model is the absorbed power law `tbabs×zpowerlw`. The best fit gives $N_\mathrm{H} = (1.054 \pm 0.009) \times 10^{22}\,\mathrm{cm^{-2}}$ and $\Gamma = 1.627 \pm 0.008$, with $\chi^2/\mathrm{dof} = 433.9/174$ (Fig. 3). The model reproduces the overall shape of the spectrum, but the residuals show a significant excess around 6 keV. The data-to-model ratio (Fig. 4) makes the feature clearer: an excess of about 10–12 per cent at 5.9–6.2 keV, at the observed-frame energy of neutral Fe Kα, $6.4/(1+z) = 6.10\,\mathrm{keV}$.

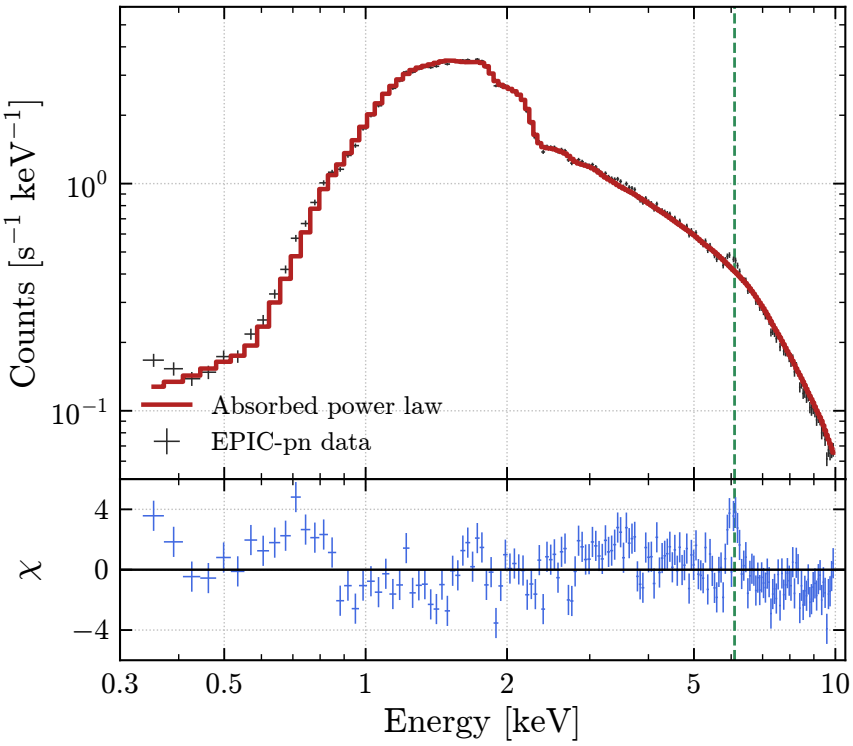


Figure 3: EPIC-pn spectrum fitted with an absorbed power law (`tbabs×zpowerlw`, $z = 0.049$) in the 0.3–10 keV band. The dashed line marks the observed-frame energy of neutral Fe Kα (6.10 keV). Lower panel: residuals in units of σ.

4.3 Absorbed power law with an Fe Kα line

A redshifted Gaussian emission line (`zgauss`) was added, giving the model `tbabs×(zpowerlw+zgauss)`. The line energy was allowed to vary between 5.8 and 7.2 keV and the width up to 0.5 keV. The fit improves to $\chi^2/\mathrm{dof} = 388.3/171$, i.e. $\Delta\chi^2 = 45.6$ for three additional free parameters; an F-test gives a probability of 3×10^{-4} that the improvement is due to chance.[1] The line has a rest-frame energy of $6.378^{+0.050}_{-0.051}\,\mathrm{keV}$, consistent with K$\alpha$ fluorescence from neutral or weakly ionised iron, and an equivalent width of $41 \pm 12\,\mathrm{eV}$. It is marginally resolved, with $\sigma = 0.124 \pm 0.055\,\mathrm{keV}$, corresponding to a full width at half maximum of $\approx$13 700 km s^{-1} (90 per cent range 7 700–19 800 km s^{-1}). The continuum parameters are almost unchanged: $N_\mathrm{H} = (1.058 \pm 0.009) \times 10^{22}\,\mathrm{cm^{-2}}$ and $\Gamma = 1.635 \pm 0.008$. The excess at

[1] The F-test is only approximate for testing the presence of an emission line (Protassov et al., 2002); given the size of $\Delta\chi^2$, the detection is nevertheless robust.

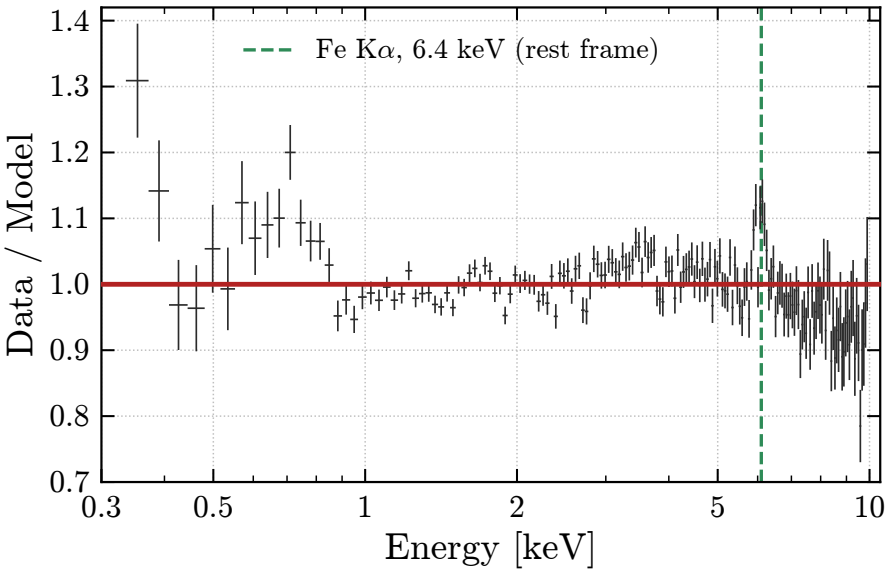


Figure 4: Data-to-model ratio for the absorbed power-law fit. The dashed line marks the expected energy of the neutral Fe Kα line (6.4 keV in the rest frame, 6.10 keV observed).

6 keV disappears from the residuals (Fig. 5). This model is adopted as the best fit. Its unfolded spectrum is shown in Fig. 6.

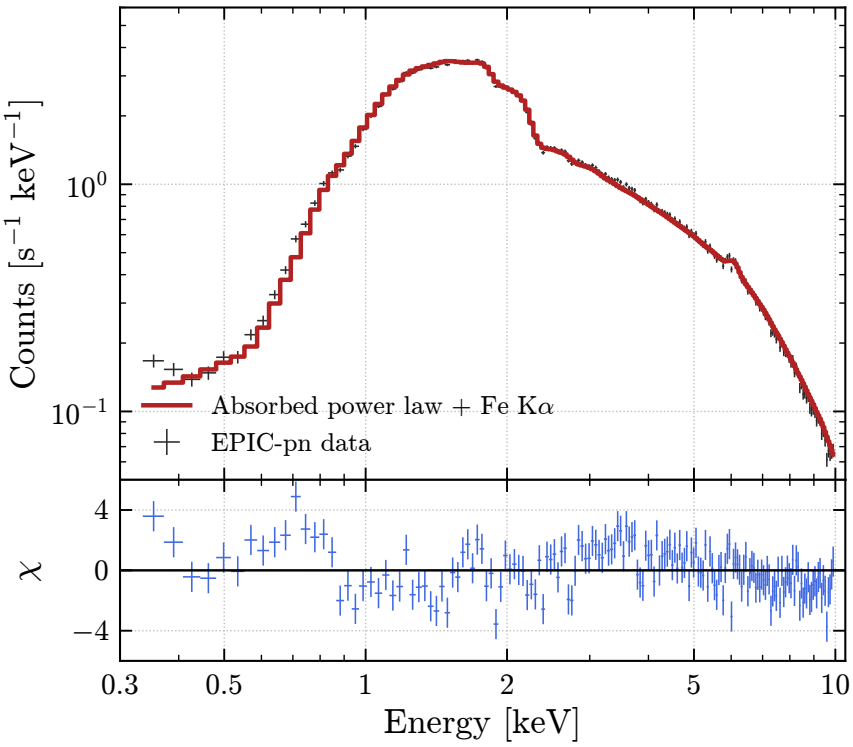


Figure 5: EPIC-pn spectrum fitted with an absorbed power law plus a Gaussian Fe Kα emission line (`tbabs`×(`zpowerlw+zgauss`)). Lower panel: residuals in units of σ.

4.4 Fluxes, luminosity and flux density

Observed fluxes were computed from the best-fitting model with 90 per cent errors from 1000 simulations of the fit covariance matrix (`flux ... err 1000 90`). Unabsorbed fluxes and

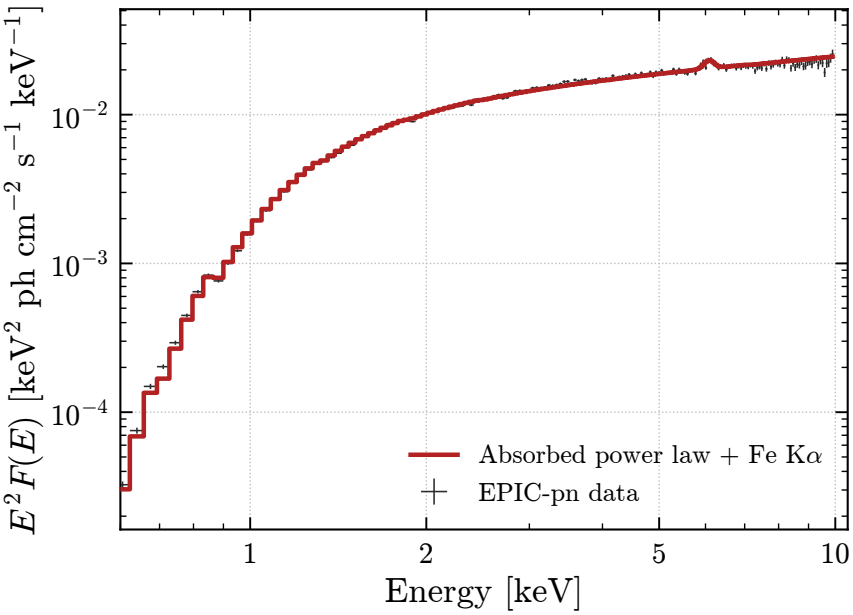


Figure 6: Unfolded $E^2F(E)$ spectrum and best-fitting absorbed power law plus Fe Kα model, shown above 0.6 keV. The rising $E^2F(E)$ reflects $\Gamma < 2$; the curvature below 2 keV is due to absorption.

rest-frame luminosities, with errors, were obtained by refitting with the convolution models `cflux` and `clumin` applied to the unabsorbed emission, `tbabs`×`cflux(zpowerlw+zgauss)` and `tbabs`×`clumin(zpowerlw+zgauss)`, so that $\log F$ and $\log L$ become fit parameters. As a cross-check, the luminosity was also computed independently as $L = 4\pi D_L^2 F_{\rm unabs}(1+z)^{\Gamma-2}$; the two agree to better than 0.5 per cent (2.70 versus 2.72×10^{44} erg s^{-1} in 2–10 keV). The monochromatic flux density at 1 keV (2.42×10^{17} Hz) was obtained from the model flux in 0.99–1.01 keV.

4.5 Best-fit parameters

Table 3 summarises all spectral parameters for both models.

Table 3: Best-fitting EPIC-pn spectral parameters of 3C 111 (0.3–10 keV, $z = 0.049$, 177 spectral bins). Errors are 90 per cent confidence ($\Delta\chi^2 = 2.706$). Luminosities are in the rest frame and corrected for absorption; the line energy is in the rest frame.

Quantity	Unit	Absorbed power law	Absorbed PL + Fe Kα
Absorption and continuum			
$N_{\rm H}$	10^{22} cm^{-2}	$1.054^{+0.009}_{-0.009}$	$1.058^{+0.009}_{-0.009}$
Γ	–	$1.627^{+0.008}_{-0.008}$	$1.635^{+0.008}_{-0.008}$
Normalisation (1 keV)	10^{-2} ph keV^{-1} cm^{-2} s^{-1}	$1.150^{+0.012}_{-0.012}$	$1.158^{+0.013}_{-0.013}$
Fe Kα emission line			
Energy (rest)	keV	–	$6.378^{+0.050}_{-0.051}$
Width σ	keV	–	$0.124^{+0.055}_{-0.055}$
Normalisation	10^{-5} ph cm^{-2} s^{-1}	–	$2.39^{+0.70}_{-0.65}$
Equivalent width	eV	–	41^{+12}_{-12}
Fit statistic			
χ^2/dof	–	433.9/174	388.3/171
χ^2_ν	–	2.49	2.27
Observed (absorbed) flux			
F (0.3–10 keV)	10^{-11} erg cm^{-2} s^{-1}	$5.286^{+0.021}_{-0.021}$	$5.282^{+0.021}_{-0.022}$
F (2–10 keV)	10^{-11} erg cm^{-2} s^{-1}	$4.594^{+0.020}_{-0.021}$	$4.590^{+0.020}_{-0.021}$
F (0.5–2 keV)	10^{-11} erg cm^{-2} s^{-1}	$0.691^{+0.003}_{-0.003}$	$0.692^{+0.003}_{-0.003}$
Unabsorbed flux			
F (0.3–10 keV)	10^{-11} erg cm^{-2} s^{-1}	7.868	$7.888^{+0.037}_{-0.036}$
F (2–10 keV)	10^{-11} erg cm^{-2} s^{-1}	4.867	$4.863^{+0.021}_{-0.021}$
Luminosity (rest frame, unabsorbed)			
L (2–10 keV)	10^{44} erg s^{-1}	2.72	$2.70^{+0.01}_{-0.01}$
L (0.3–10 keV)	10^{44} erg s^{-1}	4.39	$4.39^{+0.02}_{-0.02}$
Flux density at 1 keV			
F_ν (observed)	μJy	1.13	1.13
F_ν (unabsorbed)	μJy	7.05	7.10

4.6 Fit quality

Even for the best-fitting model the reduced χ^2 is 2.27, formally a poor fit. With 3.5×10^5 counts the statistical errors per bin are only 1–3 per cent, so residual calibration uncertainties of a few per cent are enough to raise χ^2 substantially. To locate the excess, the best-fitting model was refitted in restricted energy bands (Table 4). Above 3 keV, where absorption has no effect and $N_{\rm H}$ was therefore frozen, the model is statistically acceptable ($\chi^2_\nu = 1.02$, null-hypothesis probability 0.41) and the line energy is unchanged (6.38 keV). The excess χ^2 is thus confined to 0.3–3 keV.

Table 4: Fits of the absorbed power law + Fe Kα model in different energy bands.

Band (keV)	χ^2/dof	χ^2_ν	$N_{\rm H}$ (10^{22} cm^{-2})	Γ
0.3–10	388.3/171	2.27	1.058	1.635
0.5–10	370.7/166	2.23	1.059	1.635
3–10	108.6/106	1.02	1.059 (frozen)	1.724

The soft-band residuals (Fig. 5) are systematically positive at 0.57–0.85 keV and negative at 0.88–1.5 keV, with an additional dip near the Si K edge at 1.84 keV. A structured pattern of this kind is produced by a small shift of the energy scale rather than by a missing physical component. Ballo et al. (2011) found, for this same observation, that the pn data are affected

by *X-ray loading*: at high count rates source photons enter the on-board offset map, shifting event energies to lower values and moving events from double to single patterns. The pattern fractions of Fig. 1 show a mild deficit of doubles and excess of singles relative to the expectation, consistent with this effect, for which no reliable correction exists. Ballo et al. (2011) therefore excluded pn data below 1 keV. In the present analysis the full band is retained to constrain $N_{\rm H}$, and the hard-band parameters are regarded as the more robust.

A consequence is that the photon index depends on the band used: $\Gamma = 1.635$ in 0.3–10 keV and 1.72 in 3–10 keV. This systematic difference is ten times larger than the statistical error and should be considered the realistic uncertainty on Γ. The fluxes and luminosity are much less sensitive: they change by less than 1 per cent between models.

5 Comparison with Previous Work

Ballo et al. (2011) analysed the same observation, combining pn (1–10 keV, single events only) and MOS2 (0.4–10 keV) data, with a cut-off power law plus a weak reflection component and Fe Kα and Kβ lines. Table 5 compares their results with this work.

Table 5: Comparison with Ballo et al. (2011) for ObsID 0552180101.

Quantity	This work	Ballo et al. (2011)
$N_{\rm H}$ (10^{22} cm^{-2})	1.058 ± 0.009	1.04 ± 0.01
Γ	1.635 (0.3–10 keV), 1.72 (3–10 keV)	1.58 ± 0.01
Fe Kα energy (keV, rest)	6.38 ± 0.05	6.41 ± 0.03
Fe Kα σ (eV)	124 ± 55	89^{+53}_{-51}
Fe Kα flux (10^{-5} ph cm^{-2} s^{-1})	$2.39^{+0.70}_{-0.65}$	$2.34^{+0.59}_{-0.53}$
Fe Kα EW (eV)	41 ± 12	38^{+11}_{-9}
$F_{\rm obs}$ (0.5–2 keV; 10^{-12} cgs)	6.92	$6.83^{+0.07}_{-0.11}$
$F_{\rm obs}$ (2–10 keV; 10^{-11} cgs)	4.59	$4.83^{+0.11}_{-0.09}$
L (2–10 keV, 10^{44} erg s^{-1})	2.70	$2.75^{+0.06}_{-0.05}$

The column density, the line parameters and the soft-band flux agree within the errors. The 2–10 keV flux of this work is about 5 per cent lower, which is within typical cross-calibration differences between EPIC cameras and between event-pattern selections. The photon index depends on the continuum model, the cameras and the energy band. Ballo et al. (2011) used a power law with an exponential cut-off at 100 keV, which curves the spectrum downwards and therefore requires a flatter intrinsic index to reproduce the same observed slope; their simpler joint fit to the same XMM-Newton data gives $\Gamma = 1.61 \pm 0.01$, close to the full-band value found here. The luminosities agree to 2 per cent despite their slightly different cosmology ($H_0 = 71$).

6 Summary and Discussion

Table 6: Summary of Part C results (best-fitting model).

Spectral products	`pn_src_grp.pi`, `pn_bkg_spec.pi`, `pn_src.rmf`, `pn_src.arf`
Net count rate	8.192 ± 0.014 counts s^{-1} (livetime 42.2 ks)
Background fraction	< 1 per cent
Pile-up	none (Part A: $s = 1.003 \pm 0.004$, $d = 0.991 \pm 0.005$)
Best-fitting model	`tbabs`×`(zpowerlw+zgauss)`, 0.3–10 keV
$N_{\rm H}$	$(1.058 \pm 0.009) \times 10^{22}$ cm^{-2}
Γ	1.635 ± 0.008 (stat.), 1.63–1.72 (band-dependent)
Fe Kα	$E = 6.38 \pm 0.05$ keV, EW $= 41 \pm 12$ eV
$F_{\rm obs}$ / $F_{\rm unabs}$ (2–10 keV)	4.59 / 4.86×10^{-11} erg cm^{-2} s^{-1}
L (2–10 keV)	$(2.70 \pm 0.01) \times 10^{44}$ erg s^{-1}
F_ν (1 keV)	1.13 μJy observed, 7.10 μJy unabsorbed
Fit statistic	χ^2/dof = 388.3/171; 3–10 keV: 108.6/106

The EPIC-pn spectrum of 3C 111 is well described by a heavily absorbed, hard power law with a weak, moderately broad Fe Kα line. The main results relevant for the physical interpretation (Part D) are:

- **Absorption.** The column $N_{\rm H} \approx 1.06 \times 10^{22}$ cm^{-2} is about 3.6 times the Galactic H I column towards 3C 111 (2.9×10^{21} cm^{-2}; Kalberla et al. 2005). The line of sight crosses a Galactic molecular cloud, and the total Galactic column, including molecular gas, can reach $\sim 1.3 \times 10^{22}$ cm^{-2} (Ballo et al., 2011; Tombesi et al., 2017). The measured absorption is therefore consistent with being mostly Galactic.
- **Continuum.** $\Gamma \approx 1.6$–1.7 is flatter than the $\Gamma \approx 1.9$ typical of radio-quiet Seyfert galaxies, in agreement with previous X-ray studies of this source.
- **Fe Kα line.** The line is required at high significance but is weak (EW ≈ 40 eV), as found for other broad-line radio galaxies.
- **Luminosity.** $L_{2-10} \approx 2.7 \times 10^{44}$ erg s^{-1} places 3C 111 among luminous, radiatively efficient AGN.

Caveats and choices.

- A single absorber was used for Galactic and intrinsic absorption. With the Galactic column uncertain by several 10^{21} cm^{-2}, the pn data cannot separate the two.
- The soft band is affected by instrumental systematics (Section 4.6); the reduced χ^2 of the full-band fit reflects these rather than a missing physical component, as shown by the acceptable 3–10 keV fit. No systematic error was added to the data.
- The Fe Kβ line (7.06 keV) was not modelled separately; part of the fitted Gaussian width may be due to it.
- No reflection continuum was included, as it cannot be constrained without data above 10 keV.
- Only EPIC-pn data were used. The MOS cameras could provide an independent check of the soft-band calibration.

Products. `pn_src_spec.pi`, `pn_bkg_spec.pi`, `pn_src_grp.pi`, `pn_src.rmf`, `pn_src.arf`, the best-fit model files `best_fit_m1.xcm` and `best_fit_m2.xcm`, the parameter file `fit_results.txt`, and all figures and scripts, in `~/sas/SPASE/part_c`.

References

Ballo, L., Braito, V., Reeves, J. N., Sambruna, R. M., & Tombesi, F. 2011, MNRAS, 418, 2367 – The high-energy view of the broad-line radio galaxy 3C 111.

Kalberla, P. M. W., et al. 2005, A&A, 440, 775 – The Leiden/Argentine/Bonn (LAB) survey of Galactic H I.

Protassov, R., van Dyk, D. A., Connors, A., Kashyap, V. L., & Siemiginowska, A. 2002, ApJ, 571, 545 – Statistics, handle with care.

Tombesi, F., et al. 2017, ApJ, 842, 64 – Excess Galactic molecular absorption toward the radio galaxy 3C 111.

Verner, D. A., Ferland, G. J., Korista, K. T., & Yakovlev, D. G. 1996, ApJ, 465, 487 – Atomic data for astrophysics: photoionization cross sections.

Wilms, J., Allen, A., & McCray, R. 2000, ApJ, 542, 914 – On the absorption of X-rays in the interstellar medium.

Arnaud, K. A. 1996, ASP Conf. Ser. 101, 17 – XSPEC: the first ten years.

Gabriel, C., et al. 2004, ASP Conf. Ser. 314, 759 – The XMM-Newton SAS.

A Complete Analysis Command Log

```
# ---- environment (SAS initialised with sasenv) ----
cd ~/sas/SPASE/part_c
export SAS_CCFPATH=$HOME/sas/SPASE/CCF
export SAS_CCF=$HOME/sas/SPASE/ccf.cif
export SAS_ODF=$PWD/1683_0552180101_SCX00000SUM.SAS
mkdir -p scripts results figures/raw

# ---- missing calibration files (listed in ccf.cif, needed by rmfgen/arfgen) ----
grep -a -o "[A-Z0-9_]*_[0-9]\{4\}\.CCF" ../ccf.cif | sort -u > ../ccf_all_in_cif.txt
# download every EPN_*, XRT3_*, XMM_* file not yet present from
# https://heasarc.gsfc.nasa.gov/FTP/xmm/data/CCF/ into $SAS_CCFPATH

# ---- spectra, responses, grouping, pile-up check ----
./scripts/01_extract_spectrum.sh

# ---- check that the grouped spectrum points to its auxiliary files ----
fkeyprint pn_src_grp.pi+1 BACKFILE      # pn_bkg_spec.pi
fkeyprint pn_src_grp.pi+1 RESPFILE      # pn_src.rmf
fkeyprint pn_src_grp.pi+1 ANCRFILE      # pn_src.arf

# ---- fit, figures, table (03_make_plots.py also redraws the epatplot fractions) ----
pip install beautifultable astropy numpy matplotlib pymupdf
./scripts/run_partC.sh 2>&1 | tee results/run_partC_terminal.txt

# ---- band-dependent fit diagnostics ----
xspec <<< "@scripts/05_diagnostics.xcm" > /dev/null 2>&1
cat results/diagnostics.txt
```

B Scripts

B.1 01_extract_spectrum.sh

```
#!/bin/bash
# Part C: EPIC-pn source/background spectra, RMF, ARF and grouping
# 3C 111, XMM-Newton obs. 0552180101 (pn Prime Small Window)
set -e
cd ~/sas/SPASE/part_c
export SAS_CCFPATH=$HOME/sas/SPASE/CCF
export SAS_CCF=$HOME/sas/SPASE/ccf.cif
export SAS_ODF=$PWD/1683_0552180101_SCX00000SUM.SAS
mkdir -p figures/raw

EVT=pn_clean.fits                               # cleaned pn events (Part A)
GTI="gti($EVT:GTI00003,TIME)"                   # flare-filtered GTI, 60.3 ks
SRC="(X,Y) IN circle(27308.7,27578.4,600)"      # source, r = 30 arcsec (Part A)
BKG="(X,Y) IN circle(25689,24313,800)"          # background, r = 40 arcsec (Part A)
SEL="(CCDNR==4)&&(FLAG==0)&&(PATTERN<=4)"

rm -f pn_src_spec.pi pn_bkg_spec.pi pn_src.rmf pn_src.arf pn_src_grp.pi pn_src_all.fits

# pile-up check (all patterns)
evselect table=$EVT:EVENTS withfilteredset=yes filteredset=pn_src_all.fits destruct=yes \
  keepfilteroutput=yes expression="(CCDNR==4)&&(FLAG==0)&&($SRC)&&$GTI"
epatplot set=pn_src_all.fits plotfile=figures/raw/fig06_epatplot.ps useplotfile=yes
# SAS 22 epatplot writes figures/raw/fig06_epatplot.pdf directly; 03_make_plots.py restyles it
    ↪ as figures/fig06_pattern_fractions.pdf

# source and background spectra (standard pn channel range)
evselect table=$EVT:EVENTS withspectrumset=yes spectrumset=pn_src_spec.pi energycolumn=PI \
```

```
  spectralbinsize=5 withspecranges=yes specchannelmin=0 specchannelmax=20479 \
  expression="$SEL&&($SRC)&&$GTI"
evselect table=$EVT:EVENTS withspectrumset=yes spectrumset=pn_bkg_spec.pi energycolumn=PI \
  spectralbinsize=5 withspecranges=yes specchannelmin=0 specchannelmax=20479 \
  expression="$SEL&&($BKG)&&$GTI"

# area scaling, responses, grouping (>= 25 counts/bin, oversample 3)
backscale spectrumset=pn_src_spec.pi badpixlocation=$EVT
backscale spectrumset=pn_bkg_spec.pi badpixlocation=$EVT
rmfgen spectrumset=pn_src_spec.pi rmfset=pn_src.rmf withenergybins=yes energymin=0.1 energymax
    ↪ =12 nenergybins=2400
arfgen spectrumset=pn_src_spec.pi arfset=pn_src.arf withrmfset=yes rmfset=pn_src.rmf \
  badpixlocation=$EVT detmaptype=psf
specgroup spectrumset=pn_src_spec.pi mincounts=25 oversample=3 rmfset=pn_src.rmf \
  arfset=pn_src.arf backgndset=pn_bkg_spec.pi groupedset=pn_src_grp.pi
```

B.2 02_fit_spectrum.xcm

```
# ======================================================================
# Part C - XSPEC fit of the 3C 111 EPIC-pn spectrum (obs. 0552180101)
# Model 1: tbabs*zpowerlw            (absorbed power law)
# Model 2: tbabs*(zpowerlw+zgauss)   (+ Fe K-alpha line, 6.4 keV rest)
# Errors: 90% confidence (delta chi^2 = 2.706)
# Run from part_c/:   xspec <<< "@scripts/02_fit_spectrum.xcm"
# ======================================================================
log results/xspec_partC.log
query yes
abund wilm
xsect vern
statistic chi
cosmo 70 0 0.73
cpd /null
setplot energy

data pn_src_grp.pi
ignore bad
ignore **-0.3 10.0-**

set fh [open results/fit_results.txt w]

# ---- helper: PostScript plot + QDP data dump for matplotlib ----
proc saveplot {name cmd} {
  cpd figures/raw/$name.ps/cps
  setplot command wdata results/qdp_$name.qdp
  eval plot $cmd
  setplot delete all
  cpd /null
}

# ---- helper: parameters with 90% errors, and fit statistic ----
proc savepars {fh tag plist} {
  foreach {n name} $plist {
    set p [tcloutr param $n]
    set e [tcloutr error $n]
    puts $fh "${tag}_$name [lindex $p 0] [lindex $e 0] [lindex $e 1] [lindex $e 2]"
  }
  puts $fh "${tag}_chi2 [tcloutr stat] [lindex [tcloutr dof] 0]"
}

# ---- helper: observed fluxes (with errors), unabsorbed fluxes, luminosities ----
proc savefluxes {fh tag} {
  set bands {{0.3 10.0 0.3-10} {2.0 10.0 2-10} {0.5 2.0 0.5-2}}
```

```
  foreach b $bands {
    flux [lindex $b 0] [lindex $b 1] err 1000 90
    puts $fh "${tag}_Fobs_[lindex $b 2] [lrange [tcloutr flux 1] 0 2]"
  }
  flux 0.99 1.01
  puts $fh "${tag}_Fobs_1keV [lindex [tcloutr flux 1] 0]"
  set nH [lindex [tcloutr param 1] 0]
  newpar 1 0
  foreach b $bands {
    flux [lindex $b 0] [lindex $b 1]
    puts $fh "${tag}_Funabs_[lindex $b 2] [lindex [tcloutr flux 1] 0]"
  }
  flux 0.99 1.01
  puts $fh "${tag}_Funabs_1keV [lindex [tcloutr flux 1] 0]"
  lumin 2.0 10.0 0.049
  puts $fh "${tag}_Lxspec_2-10 [lindex [tcloutr lumin 1] 0]"
  lumin 0.3 10.0 0.049
  puts $fh "${tag}_Lxspec_0.3-10 [lindex [tcloutr lumin 1] 0]"
  newpar 1 $nH
}

# ================= Plot 1: background-subtracted spectrum =================
saveplot fig01_spectrum ldata

# ================= Model 1: absorbed power law =================
model tbabs*zpowerlw & /*
newpar 1 0.9
newpar 2 1.7
newpar 3 0.049
freeze 3
newpar 4 0.01
fit 200
fit 200
error maximum 10.0 2.706 1 2 4
savepars $fh m1 {1 nH 2 Gamma 4 norm}
saveplot fig02_fit_m1 "ldata delchi"
saveplot fig03_ratio_m1 ratio
save all results/best_fit_m1.xcm
savefluxes $fh m1

# ================= Model 2: absorbed power law + Fe K-alpha =================
set nH1 [lindex [tcloutr param 1] 0]
set G1  [lindex [tcloutr param 2] 0]
set K1  [lindex [tcloutr param 4] 0]
model tbabs*(zpowerlw+zgauss) & /*
newpar 1 $nH1
newpar 2 $G1
newpar 3 0.049
freeze 3
newpar 4 $K1
newpar 5 6.4 0.01 5.8 5.8 7.2 7.2
newpar 6 0.05 0.01 0.0 0.0 0.5 0.5
newpar 7 0.049
freeze 7
newpar 8 5e-5 1e-6 0 0 1e-2 1e-2
fit 200
fit 200
error maximum 10.0 2.706 1 2 4 5 6 8
savepars $fh m2 {1 nH 2 Gamma 4 norm 5 LineE 6 Sigma 8 LineNorm}
eqwidth 3 err 1000 90
puts $fh "m2_EW [lrange [tcloutr eqwidth 1] 0 2]"
saveplot fig04_fit_m2 "ldata delchi"
saveplot fig05_unfolded_m2 eeufspec
```

```
save all results/best_fit_m2.xcm
savefluxes $fh m2

# ===== Model 2: unabsorbed flux (cflux) and luminosity (clumin) with errors =====
set nH2 [lindex [tcloutr param 1] 0]
set G2  [lindex [tcloutr param 2] 0]
set K2  [lindex [tcloutr param 4] 0]
set E2  [lindex [tcloutr param 5] 0]
set S2  [lindex [tcloutr param 6] 0]
set N2  [lindex [tcloutr param 8] 0]
foreach b {{2.0 10.0 2-10} {0.3 10.0 0.3-10}} {
  model tbabs*cflux(zpowerlw+zgauss) & /*
  newpar 1 $nH2
  newpar 2 [lindex $b 0]
  newpar 3 [lindex $b 1]
  newpar 4 -10.3
  newpar 5 $G2
  newpar 6 0.049
  newpar 7 $K2
  freeze 7
  newpar 8 $E2
  newpar 9 $S2
  newpar 10 0.049
  newpar 11 $N2
  fit 200
  error maximum 10.0 2.706 4
  puts $fh "m2_cflux_[lindex $b 2] [lindex [tcloutr param 4] 0] [lrange [tcloutr error 4] 0 1]"

  model tbabs*clumin(zpowerlw+zgauss) & /*
  newpar 1 $nH2
  newpar 2 [lindex $b 0]
  newpar 3 [lindex $b 1]
  newpar 4 0.049
  newpar 5 44.5
  newpar 6 $G2
  newpar 7 0.049
  newpar 8 $K2
  freeze 8
  newpar 9 $E2
  newpar 10 $S2
  newpar 11 0.049
  newpar 12 $N2
  fit 200
  error maximum 10.0 2.706 5
  puts $fh "m2_clumin_[lindex $b 2] [lindex [tcloutr param 5] 0] [lrange [tcloutr error 5] 0 1]"
}

close $fh
log none
exit
```

B.3 05_diagnostics.xcm

```
# Part C diagnostics: where does the excess chi^2 come from? (Model 2)
query yes
log results/xspec_diagnostics.log
@results/best_fit_m2.xcm
set fh [open results/diagnostics.txt w]
fit 200
puts $fh "band_0.3-10 chi2=[tcloutr stat] dof=[lindex [tcloutr dof] 0] nH=[lindex [tcloutr param
    ↪ 1] 0] Gamma=[lindex [tcloutr param 2] 0]"
ignore **-0.5
```

```
fit 200
puts $fh "band_0.5-10 chi2=[tcloutr stat] dof=[lindex [tcloutr dof] 0] nH=[lindex [tcloutr param
    ↪  1] 0] Gamma=[lindex [tcloutr param 2] 0]"
ignore **-3.0
freeze 1
fit 200
puts $fh "band_3-10(nH frozen) chi2=[tcloutr stat] dof=[lindex [tcloutr dof] 0] nH=[lindex [
    ↪ tcloutr param 1] 0] Gamma=[lindex [tcloutr param 2] 0] LineE=[lindex [tcloutr param 5] 0]
    ↪ "
close $fh
log none
exit
```

B.4 run_partC.sh

```
#!/bin/bash
# Part C runner: XSPEC fit -> PDF figures -> parameter table
cd ~/sas/SPASE/part_c || exit 1
mkdir -p results figures/raw
rm -f results/xspec_partC.log results/fit_results.txt results/best_fit_m*.xcm \
      results/qdp_*.qdp figures/raw/fig0[1-5]_*.ps figures/raw/fig0[1-5]_*.pdf
xspec <<< "@scripts/02_fit_spectrum.xcm"
for f in figures/raw/fig0[1-5]_*.ps; do ps2pdf "$f" "${f%.ps}.pdf"; done
python scripts/03_make_plots.py
python scripts/04_results_table.py
```

B.5 03_make_plots.py

```
#!/usr/bin/env python3
"""
Part C: publication-quality figures (PDF + PNG) for the 3C 111 EPIC-pn spectral analysis.

Inputs
  results/qdp_fig0[1-5]_*.qdp      XSPEC 'wdata' dumps written by 02_fit_spectrum.xcm
  figures/raw/fig06_epatplot.pdf   epatplot output (pattern fractions are read from its vector
    ↪ paths)
Outputs
  figures/fig01 ... fig06 (.pdf and .png), results/epatplot_fractions.csv

Style: Computer Modern fonts, large labels, inward ticks on all four sides,
light dotted grid, thick model lines, units in square brackets, no titles.
"""
from pathlib import Path
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from matplotlib.ticker import NullFormatter, FixedLocator, MultipleLocator

RES, FIG = Path("results"), Path("figures")
FIG.mkdir(exist_ok=True)
Z = 0.049
E_FE_OBS = 6.4 / (1 + Z)                    # Fe K-alpha, 6.4 keV rest -> 6.10 keV observed

# ---------------- house style ----------------
RED, ORANGE, GREEN, BLUE = "#B22222", "#FF8C00", "#2E8B57", "#4169E1"
DATA = "#2B2B2B"
plt.rcParams.update({
    "font.family": "serif", "font.serif": ["cmr10", "Computer Modern Roman", "DejaVu Serif"],
    "mathtext.fontset": "cm", "axes.formatter.use_mathtext": True,
```

```
    "font.size": 17, "axes.labelsize": 20, "xtick.labelsize": 17, "ytick.labelsize": 17,
    "legend.fontsize": 15, "legend.frameon": False, "legend.handlelength": 1.8,
    "axes.linewidth": 1.2, "axes.unicode_minus": False,
    "xtick.direction": "in", "ytick.direction": "in", "xtick.top": True, "ytick.right": True,
    "xtick.major.size": 7, "ytick.major.size": 7, "xtick.minor.size": 3.5, "ytick.minor.size":
    ↪ 3.5,
    "xtick.major.width": 1.1, "ytick.major.width": 1.1, "xtick.minor.width": 0.8, "ytick.minor.
    ↪ width": 0.8,
    "xtick.major.pad": 7, "ytick.major.pad": 6,
    "axes.grid": True, "grid.color": "0.75", "grid.linestyle": ":", "grid.linewidth": 0.8,
    "axes.axisbelow": True,
    "lines.linewidth": 2.6, "errorbar.capsize": 0,
    "savefig.bbox": "tight", "savefig.dpi": 300, "pdf.fonttype": 42, "ps.fonttype": 42})

E_LABEL = r"Energy [keV]"
COUNTS = r"Counts [s$^{-1}$ keV$^{-1}$]"


# ---------------- helpers ----------------
def read_qdp(path):
    """List of arrays, one per plot panel (panels separated by 'NO' lines)."""
    blocks, cur = [], []
    for line in open(path):
        s = line.strip()
        if not s or s[0] in "!@" or s.upper().startswith(("READ", "SKIP")):
            continue
        if s.upper().startswith("NO"):
            if cur:
                blocks.append(cur); cur = []
            continue
        row = []
        for t in s.split():
            try:
                row.append(float(t))
            except ValueError:
                row.append(np.nan)
        cur.append(row)
    if cur:
        blocks.append(cur)
    out = []
    for b in blocks:
        n = min(len(r) for r in b)
        out.append(np.array([r[:n] for r in b]))
    return out


def energy_axis(ax, emin=0.3, emax=10.5, label=True):
    ax.set_xscale("log")
    ax.set_xlim(emin, emax)
    ticks = [t for t in (0.3, 0.5, 1, 2, 5, 10) if emin <= t <= emax]
    ax.xaxis.set_major_locator(FixedLocator(ticks))
    ax.set_xticklabels([f"{t:g}" for t in ticks])
    ax.xaxis.set_minor_formatter(NullFormatter())
    if label:
        ax.set_xlabel(E_LABEL)


def save(fig, name):
    fig.savefig(FIG / f"{name}.pdf")
    fig.savefig(FIG / f"{name}.png")
    plt.close(fig)
    print("wrote", FIG / f"{name}.pdf")
```

```
def data_points(ax, d, label="EPIC-pn data", color=DATA):
    ax.errorbar(d[:, 0], d[:, 2], xerr=d[:, 1], yerr=d[:, 3], fmt="none",
                ecolor=color, elinewidth=0.9, label=label, zorder=2)


def model_line(ax, d, label, color=RED):
    if d.shape[1] > 4 and np.isfinite(d[:, 4]).any():
        ax.plot(d[:, 0], d[:, 4], color=color, drawstyle="steps-mid", label=label, zorder=3)


def residual_panel(ax, r):
    ax.errorbar(r[:, 0], r[:, 2], xerr=r[:, 1], yerr=r[:, 3], fmt="none",
                ecolor=BLUE, elinewidth=0.9, zorder=2)
    ax.axhline(0, color="black", lw=1.4, zorder=3)
    ax.set_ylim(-6, 6)
    ax.yaxis.set_major_locator(FixedLocator([-4, 0, 4]))
    ax.set_ylabel(r"$\chi$")


# ---------------- Fig. 1: spectrum ----------------
def fig_spectrum():
    d = read_qdp(RES / "qdp_fig01_spectrum.qdp")[0]
    fig, ax = plt.subplots(figsize=(7.5, 5.4))
    data_points(ax, d)
    ax.set_yscale("log")
    energy_axis(ax)
    ax.set_ylabel(COUNTS)
    ax.set_ylim(0.05, 6)
    save(fig, "fig01_spectrum")


# ---------------- Figs. 2 and 4: fit + residuals ----------------
def fig_fit(qdp, model_label, name, mark_fe=False):
    d, r = read_qdp(RES / qdp)[:2]
    fig, (a1, a2) = plt.subplots(2, 1, sharex=True, figsize=(7.5, 6.8),
                                 gridspec_kw=dict(height_ratios=[3, 1.15], hspace=0))
    data_points(a1, d)
    model_line(a1, d, model_label)
    a1.set_yscale("log")
    a1.set_ylim(0.05, 6)
    a1.set_ylabel(COUNTS)
    a1.legend(loc="lower left")
    residual_panel(a2, r)
    if mark_fe:
        for a in (a1, a2):
            a.axvline(E_FE_OBS, ls="--", color=GREEN, lw=1.6, zorder=1)
    energy_axis(a2)
    fig.align_ylabels((a1, a2))
    save(fig, name)


# ---------------- Fig. 3: ratio ----------------
def fig_ratio():
    r = read_qdp(RES / "qdp_fig03_ratio_m1.qdp")[0]
    fig, ax = plt.subplots(figsize=(7.5, 4.6))
    ax.errorbar(r[:, 0], r[:, 2], xerr=r[:, 1], yerr=r[:, 3], fmt="none",
                ecolor=DATA, elinewidth=0.9, zorder=2)
    ax.axhline(1, color=RED, zorder=3)
    ax.axvline(E_FE_OBS, ls="--", color=GREEN, lw=1.8, zorder=1,
               label=r"Fe K$\alpha$, 6.4 keV (rest frame)")
    energy_axis(ax)
    ax.set_ylim(0.7, 1.42)
```

```
    ax.yaxis.set_major_locator(MultipleLocator(0.1))
    ax.set_ylabel(r"Data / Model")
    ax.legend(loc="upper left", bbox_to_anchor=(0.15, 0.99))
    save(fig, "fig03_ratio_absorbed_powerlaw")


# ---------------- Fig. 5: unfolded spectrum ----------------
def fig_unfolded():
    d = read_qdp(RES / "qdp_fig05_unfolded_m2.qdp")[0]
    sel = d[:, 0] >= 0.6
    d = d[sel]
    fig, ax = plt.subplots(figsize=(7.5, 5.4))
    data_points(ax, d)
    model_line(ax, d, r"Absorbed power law + Fe K$\alpha$")
    ax.set_yscale("log")
    ax.set_ylim(0.5 * np.nanmin(d[:, 2]), 2.0 * np.nanmax(d[:, 2]))
    energy_axis(ax, emin=0.6)
    ax.set_ylabel(r"$E^{2}F(E)$ [keV$^{2}$ ph cm$^{-2}$ s$^{-1}$ keV$^{-1}$]")
    ax.legend(loc="lower right")
    save(fig, "fig05_unfolded_spectrum")


# ---------------- Fig. 6: pattern fractions (from epatplot) ----------------
def read_epatplot(pdf=FIG / "raw" / "fig06_epatplot.pdf"):
    """Recover observed and model single/double fractions from page 2 of the epatplot PDF.

    The axes are calibrated from the major tick marks (x: 10^2, 10^3, 10^4 eV; y: 0.0 ... 1.0).
    Colours used by epatplot: singles = purple, doubles = blue.
    """
    import pymupdf
    page = pymupdf.open(pdf)[1]
    draws = page.get_drawings()
    black = [g for g in draws if g.get("color") == (0.0, 0.0, 0.0) and g["type"] == "fs"]
    xt = sorted({round(g["rect"].x0, 3) for g in black
                 if abs(g["rect"].height - 3.5) < 0.05 and g["rect"].width < 0.01})
    yt = sorted({round(g["rect"].y0, 3) for g in black
                 if abs(g["rect"].width - 3.5) < 0.05 and g["rect"].height < 0.01}, reverse=True
    ↪ )
    x0, x1 = xt[0], xt[1]                          # 10^2 and 10^3 eV
    y0, y1 = yt[0], yt[-1]                         # 0.0 and 1.0

    def to_data(path):
        pts = []
        for it in path["items"]:
            if it[0] == "l":
                pts += [it[1], it[2]]
        xy = np.array([[10 ** (2 + (p.x - x0) / (x1 - x0)) / 1e3, (p.y - y0) / (y1 - y0)] for p
    ↪ in pts])
        keep = [0] + [k for k in range(1, len(xy)) if not np.allclose(xy[k], xy[k - 1])]
        return xy[keep]

    def colour(g, rgb):
        return g["type"] == "s" and g.get("color") is not None and \
            np.allclose(g["color"], rgb, atol=0.01)

    purple, blue = (0.478, 0.408, 0.651), (0.204, 0.541, 0.741)
    paths = {"s": [to_data(g) for g in draws if colour(g, purple)],
             "d": [to_data(g) for g in draws if colour(g, blue)]}
    out = {}
    for k, lst in paths.items():
        obs = max(lst, key=len)                    # histogram has the most vertices
        mod = min(lst, key=len)                    # smooth model curve
        lo, hi, v = [], [], []
```

```
        for i in range(len(obs) - 1):
            if abs(obs[i, 1] - obs[i + 1, 1]) < 1e-9 and obs[i + 1, 0] > obs[i, 0]:
                lo.append(obs[i, 0]); hi.append(obs[i + 1, 0]); v.append(obs[i, 1])
        out[k] = dict(lo=np.array(lo), hi=np.array(hi), obs=np.array(v), model=mod)
    return out


def fig_pattern():
    try:
        p = read_epatplot()
    except Exception as err:                        # pymupdf missing or file absent
        print("SKIPPED fig06:", err)
        return
    s, d = p["s"], p["d"]
    lo, hi = s["lo"], s["hi"]
    ok = lo >= 0.33                                  # first bins (< 0.34 keV) are noise-dominated
    lo, hi = lo[ok], hi[ok]
    so, do = s["obs"][ok], d["obs"][ok]
    ec = np.sqrt(lo * hi)

    def interp(m, x):
        return np.interp(np.log(x), np.log(m[:, 0]), m[:, 1])

    # the clean event list keeps only PATTERN<=4, so compare with the model renormalised to s+d
    egrid = np.logspace(np.log10(0.25), np.log10(15), 400)
    ms, md = interp(s["model"], egrid), interp(d["model"], egrid)
    msn, mdn = ms / (ms + md), md / (ms + md)
    ms_c, md_c = interp(s["model"], ec), interp(d["model"], ec)
    rs, rd = so / (ms_c / (ms_c + md_c)), do / (md_c / (ms_c + md_c))

    band = (ec >= 0.5) & (ec <= 2.0)
    w = (hi - lo)[band]
    print("0.5-2 keV observed/model (renormalised): singles %.3f, doubles %.3f"
          % (np.sum(so[band] * w) / np.sum((so[band] / rs[band]) * w),
             np.sum(do[band] * w) / np.sum((do[band] / rd[band]) * w)))
    np.savetxt(RES / "epatplot_fractions.csv",
               np.column_stack([lo, hi, so, do, ms_c / (ms_c + md_c), md_c / (ms_c + md_c)]),
               delimiter=",", fmt="%.6g",
               header="E_lo_keV,E_hi_keV,single_obs,double_obs,single_model,double_model")

    fig, (a1, a2) = plt.subplots(2, 1, sharex=True, figsize=(7.5, 6.8),
                                 gridspec_kw=dict(height_ratios=[2.2, 1], hspace=0))
    for a in (a1, a2):
        a.axvspan(0.5, 2.0, color="0.92", zorder=0)
    edges = np.append(lo, hi[-1])
    a1.stairs(so, edges, color=RED, lw=1.1, alpha=0.9, baseline=None, label="Singles (observed)"
    ↪ )
    a1.stairs(do, edges, color=BLUE, lw=1.1, alpha=0.9, baseline=None, label="Doubles (observed)
    ↪ ")
    a1.plot(egrid, msn, color="black", lw=2.2, label="Expected (no pile-up)")
    a1.plot(egrid, mdn, color="black", lw=2.2)
    a1.set_ylim(0, 1)
    a1.yaxis.set_major_locator(MultipleLocator(0.2))
    a1.set_ylabel(r"Pattern fraction")
    a1.legend(loc="center left", bbox_to_anchor=(0.01, 0.49), fontsize=14)

    a2.plot(ec, rs, ".", color=RED, ms=3.5)
    a2.plot(ec, rd, ".", color=BLUE, ms=3.5)
    a2.axhline(1, color="black", lw=1.4)
    a2.set_ylim(0.8, 1.2)
    a2.yaxis.set_major_locator(FixedLocator([0.9, 1.0, 1.1]))
    a2.set_ylabel(r"Obs. / Exp.")
    energy_axis(a2, emin=0.3, emax=15)
```

```
    a2.xaxis.set_major_locator(FixedLocator([0.3, 0.5, 1, 2, 5, 10]))
    a2.set_xticklabels(["0.3", "0.5", "1", "2", "5", "10"])
    fig.align_ylabels((a1, a2))
    save(fig, "fig06_pattern_fractions")


for job in (fig_spectrum,
            lambda: fig_fit("qdp_fig02_fit_m1.qdp", "Absorbed power law",
                            "fig02_fit_absorbed_powerlaw", mark_fe=True),
            fig_ratio,
            lambda: fig_fit("qdp_fig04_fit_m2.qdp", r"Absorbed power law + Fe K$\alpha$",
                            "fig04_fit_powerlaw_FeKa"),
            fig_unfolded,
            fig_pattern):
    try:
        job()
    except (FileNotFoundError, IndexError) as err:
        print("SKIPPED:", err)
```

B.6 04_results_table.py

```
#!/usr/bin/env python3
"""Part C: best-fit parameter table (Markdown + LaTeX) from results/fit_results.txt."""
from pathlib import Path
import numpy as np
from astropy.cosmology import FlatLambdaCDM
import astropy.units as u

Z = 0.049
COSMO = FlatLambdaCDM(H0=70, Om0=0.27)     # same as XSPEC 'cosmo 70 0 0.73'
KEV_HZ = 2.417989e17                         # 1 keV in Hz
RES = Path("results")
DL = COSMO.luminosity_distance(Z).to(u.cm).value

d = {}
for line in open(RES / "fit_results.txt"):
    t = line.split()
    if len(t) < 2:
        continue
    v = []
    for x in t[1:]:
        try:
            v.append(float(x))
        except ValueError:
            v.append(x)
    d[t[0]] = v

NONE = ("-", "--")


def fmt(v, lo=None, hi=None, nd=3):
    """Value with asymmetric 90% errors (lo/hi are absolute confidence bounds)."""
    if v is None:
        return NONE
    if lo is None or hi is None or (lo == 0 and hi == 0):
        return f"{v:.{nd}f}", f"${v:.{nd}f}$"
    up, dn = hi - v, v - lo
    return (f"{v:.{nd}f} (+{up:.{nd}f}/-{dn:.{nd}f})",
            f"${v:.{nd}f}^{{+{up:.{nd}f}}}_{{-{dn:.{nd}f}}}$")


def par(tag, name, scale=1.0, nd=3):
```

```
    k = f"{tag}_{name}"
    if k not in d:
        return NONE
    v, lo, hi = (d[k] + [0, 0])[:3]
    return fmt(v * scale, lo * scale, hi * scale, nd)


def lgpar(key, scale=1.0, nd=3):
    if key not in d:
        return NONE
    v, lo, hi = d[key][:3]
    if lo == 0 and hi == 0:
        return fmt(10**v * scale, nd=nd)
    return fmt(10**v * scale, 10**lo * scale, 10**hi * scale, nd)


def chi(tag):
    k = f"{tag}_chi2"
    if k not in d:
        return NONE, NONE
    c, dof = d[k][:2]
    return (f"{c:.1f}/{int(dof)}", f"${c:.1f}/{int(dof)}$"), (f"{c/dof:.2f}", f"${c/dof:.2f}$")


def fnu(tag, kind):
    k = f"{tag}_{kind}_1keV"
    if k not in d:
        return NONE
    F = d[k][0] / 0.02                                  # erg cm^-2 s^-1 keV^-1
    return fmt(F / KEV_HZ / 1e-29, nd=2)                # micro-Jansky


def lum_est(tag, band):
    """Rest-frame luminosity from observed-band unabsorbed flux, K-corrected."""
    k, g = f"{tag}_Funabs_{band}", f"{tag}_Gamma"
    if k not in d or g not in d:
        return None
    return 4 * np.pi * DL**2 * d[k][0] * (1 + Z)**(d[g][0] - 2)


def lum(tag, band):
    L = lum_est(tag, band)
    return NONE if L is None else fmt(L / 1e44, nd=2)


def unabs_m2(band):
    return lgpar(f"m2_cflux_{band}", 1e11) if f"m2_cflux_{band}" in d else par("m2", f"Funabs_{
    ↪ band}", 1e11)


def lum_m2(band):
    return lgpar(f"m2_clumin_{band}", 1e-44, 2) if f"m2_clumin_{band}" in d else lum("m2", band)


c1, r1 = chi("m1")
c2, r2 = chi("m2")
U_F = ("10^-11 erg cm^-2 s^-1", r"$10^{-11}$ erg cm$^{-2}$ s$^{-1}$")
rows = [
    ("N_H", r"$N_{\rm H}$", "10^22 cm^-2", r"$10^{22}$ cm$^{-2}$", par("m1", "nH"), par("m2", "
    ↪ nH")),
    ("Photon index Γ", r"$\Gamma$", "-", "--", par("m1", "Gamma"), par("m2", "Gamma")),
    ("PL norm (1 keV)", r"PL norm (1 keV)", "10^-2 ph keV^-1 cm^-2 s^-1",
```

```
     r"$10^{-2}$ ph keV$^{-1}$ cm$^{-2}$ s$^{-1}$", par("m1", "norm", 100), par("m2", "norm",
    ↪ 100)),
    ("Fe Kα energy (rest)", r"Fe K$\alpha$ $E$ (rest)", "keV", "keV", NONE, par("m2", "LineE")),
    ("Fe Kα σ", r"Fe K$\alpha$ $\sigma$", "keV", "keV", NONE, par("m2", "Sigma")),
    ("Fe Kα norm", r"Fe K$\alpha$ norm", "10^-5 ph cm^-2 s^-1", r"$10^{-5}$ ph cm$^{-2}$ s$^{-1}
    ↪ $",
     NONE, par("m2", "LineNorm", 1e5, 2)),
    ("Fe Kα EW (obs.)", r"Fe K$\alpha$ EW (obs.)", "eV", "eV", NONE, par("m2", "EW", 1e3, 0)),
    ("χ²/dof", r"$\chi^2$/dof", "-", "--", c1, c2),
    ("Reduced χ²", r"$\chi^2_\nu$", "-", "--", r1, r2),
    ("F_obs (0.3-10 keV)", r"$F_{\rm obs}$ (0.3--10 keV)", *U_F, par("m1", "Fobs_0.3-10", 1e11),
    ↪  par("m2", "Fobs_0.3-10", 1e11)),
    ("F_obs (2-10 keV)", r"$F_{\rm obs}$ (2--10 keV)", *U_F, par("m1", "Fobs_2-10", 1e11), par("
    ↪ m2", "Fobs_2-10", 1e11)),
    ("F_obs (0.5-2 keV)", r"$F_{\rm obs}$ (0.5--2 keV)", *U_F, par("m1", "Fobs_0.5-2", 1e11),
    ↪ par("m2", "Fobs_0.5-2", 1e11)),
    ("F_unabs (0.3-10 keV)", r"$F_{\rm unabs}$ (0.3--10 keV)", *U_F, par("m1", "Funabs_0.3-10",
    ↪ 1e11), unabs_m2("0.3-10")),
    ("F_unabs (2-10 keV)", r"$F_{\rm unabs}$ (2--10 keV)", *U_F, par("m1", "Funabs_2-10", 1e11),
    ↪  unabs_m2("2-10")),
    ("L (2-10 keV, rest)", r"$L$ (2--10 keV, rest)", "10^44 erg s^-1", r"$10^{44}$ erg s$^{-1}$"
    ↪ , lum("m1", "2-10"), lum_m2("2-10")),
    ("L (0.3-10 keV, rest)", r"$L$ (0.3--10 keV, rest)", "10^44 erg s^-1", r"$10^{44}$ erg s$
    ↪ ^{-1}$", lum("m1", "0.3-10"), lum_m2("0.3-10")),
    ("F_ν (1 keV, observed)", r"$F_\nu$ (1 keV, observed)", "μJy", r"$\mu$Jy", fnu("m1", "Fobs")
    ↪ , fnu("m2", "Fobs")),
    ("F_ν (1 keV, unabsorbed)", r"$F_\nu$ (1 keV, unabsorbed)", "μJy", r"$\mu$Jy", fnu("m1", "
    ↪ Funabs"), fnu("m2", "Funabs")),
]

H1, H2 = "Absorbed power law", "Absorbed power law + Fe Kα"
md = [f"| Quantity | Unit | {H1} | {H2} |", "|---|---|---|---|"]
md += [f"| {qm} | {um} | {a[0]} | {b[0]} |" for qm, qt, um, ut, a, b in rows]
(RES / "best_fit_table.md").write_text("\n".join(md) + "\n")

tex = [r"\begin{table}", r"\centering",
       r"\caption{Best-fitting EPIC-pn spectral parameters of 3C\,111 (0.3--10 keV, $z=0.049$).
    ↪ "
       r"Errors are 90 per cent confidence ($\Delta\chi^2=2.706$).}",
       r"\begin{tabular}{llcc}", r"\hline",
       r"Quantity & Unit & Absorbed power law & Absorbed power law + Fe K$\alpha$ \\", r"\hline"
    ↪ ]
tex += [f"{qt} & {ut} & {a[1]} & {b[1]} \\\\" for qm, qt, um, ut, a, b in rows]
tex += [r"\hline", r"\end{tabular}", r"\end{table}"]
(RES / "best_fit_table.tex").write_text("\n".join(tex) + "\n")

print("\n".join(md))
print(f"\nD_L = {DL/3.0857e24:.1f} Mpc")
print("\nwrote results/best_fit_table.md and results/best_fit_table.tex")
```